

Threat Analysis Tools

CTI Foundations, File and Hash Enrichment, IP and Domain Intelligence

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Part 1	Part 2	Part 3	Part 4
CTI Concepts and Frameworks	File Enrichment and Sandboxing	IP and Domain OSINT	Artefact Investigation

Project Overview

This document covers the Threat Analysis Tools learning path on TryHackMe, a four-part series that builds practical threat intelligence skills from foundational concepts through to hands-on enrichment of file, hash, IP, and domain artefacts.

The path progresses logically: Part 1 establishes the theory, terminology, and frameworks that define how threat intelligence is structured and shared. Parts 2 and 3 apply that theory directly to the two most common artefact types encountered in SOC work. Part 4 is a capstone exercise requiring the analyst to extract and distil intelligence from a set of flagged artefacts independently.

Project Objectives

- Understand the three levels of threat data (Data, Information, Intelligence) and the four classifications of CTI (Strategic, Tactical, Operational, Technical).
- Learn the six-phase CTI lifecycle (Direction, Collection, Processing, Analysis, Dissemination, Feedback) and apply it to a real investigation scenario.
- Understand the key CTI frameworks and standards: MITRE ATT&CK, TAXII, STIX, and the Cyber Kill Chain.
- Detect file masquerading techniques including misleading names and double extensions, and extract hashes using PowerShell and CMD.
- Enrich file and hash artefacts using Virus Total (detection, behavior, and MITRE mapping), MalwareBazaar (vendor verdict comparison), and Hybrid Analysis (sandbox process tree and network analysis).
- Enrich IP and domain artefacts using RDAP and WHOIS for ownership, Shodan and Censys for exposed services, crt.sh for certificate transparency, and Virus Total and Cisco Talos for reputation.
- Understand DNS record types (A, MX, TXT, CNAME) and their role in identifying attacker infrastructure and typo squatted domains.
- Apply multi-source correlation to build a complete threat profile from individual artefacts and produce evidence-driven SOC recommendations.

Tools Used

Tool, Platform	Purpose in This Lab
Virus Total	Checks file hashes, URLs, IPs, and domains against 70+ antivirus vendors. Also provides behaviour analysis, MITRE ATT&CK technique mapping, passive DNS history, and community comments.

MalwareBazaar	Abuse.ch malware repository. Allows hash-based lookups to see how different vendors classify a specific sample. Search syntax: sha256:hash. Useful for cross-referencing VirusTotal verdicts.
Hybrid Analysis	Free online sandbox. Runs files in an isolated environment and returns a full report: static indicators, process execution tree, network connections, dropped files, and assigned tags.
PowerShell, Get-FileHash	Generates MD5 and SHA256 hashes on Windows. Used to produce the hash fingerprint needed for threat intelligence platform lookups.
RDAP, client.rdap.org	Replacement for legacy WHOIS. Provides structured JSON-based registration data for IP ranges and domains including registration date, country, and assigned roles (administrative, technical).
Shodan	Search engine for internet-exposed devices. Returns open ports, running services, banners, and known vulnerabilities for a given IP address.
Censys	Internet scanning platform with detailed TLS certificate data. Used to retrieve the SHA256 fingerprint of a certificate hosted on a suspicious IP.
crt.sh	Certificate Transparency log search. Used to find all SSL/TLS certificates ever issued for a domain or certificate fingerprint. Reveals subdomains and attacker infrastructure reuse.
Cisco Talos Intelligence	IP and domain reputation database from Cisco. Provides a historical reputation score, email reputation, and threat category for investigated infrastructure.

Intro to Cyber Threat Intelligence

Cyber Threat Intelligence (CTI) is evidence-based knowledge about adversaries, including their indicators, tactics, motivations, and actionable recommendations. It moves raw data from an alert into a structured, contextualized body of knowledge that helps teams make faster and better defensive decisions.

Data vs Information vs Intelligence

Level	Definition	Example
Data	Discrete, raw artefacts with no context.	IP address 34.253.159.159, file hash, malicious URL.
Information	Multiple data points combined to answer a question.	How many times did employees access an unknown domain this month?
Intelligence	Correlated data and information analyzed to identify patterns and support decisions.	This IP is associated with Emotet C2 infrastructure based on behavior patterns and community reports.

CTI Classifications

- Strategic Intel: High-level briefings for senior leadership about threat actor motivations, geopolitical risk, and long-term trends. Not technical.
- Tactical Intel: Details about adversary TTPs (Tactics, Techniques, and Procedures) to inform defensive tool configurations and detection rules.
- Operational Intel: Information about specific active campaigns, threat actor infrastructure, and planned attacks.

- **Technical Intel:** Specific artefacts such as IP addresses, file hashes, domains, and URLs. These feed directly into SIEM rules, firewall blocklists, and threat intelligence platforms.

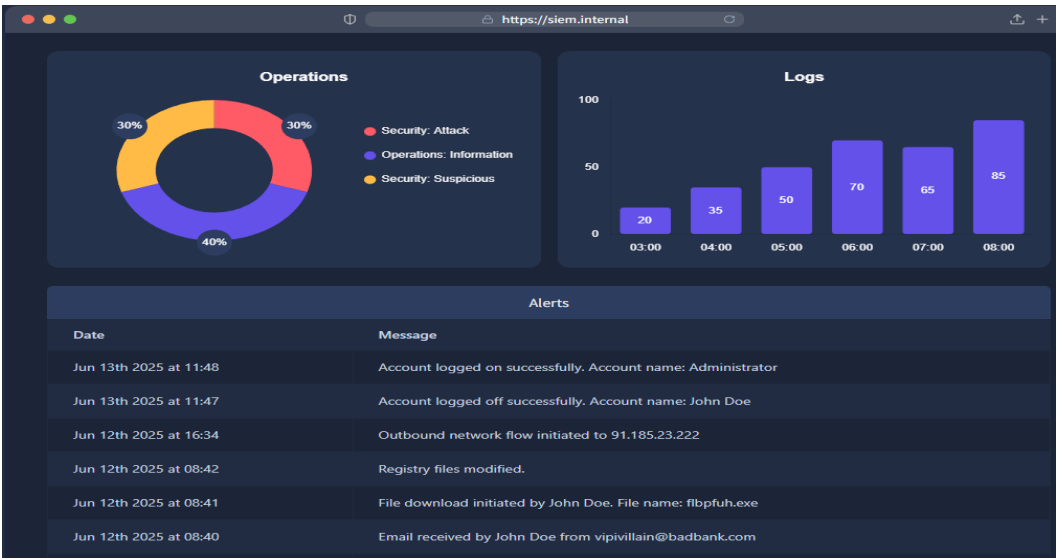
The CTI Lifecycle

Phase	What Happens
Direction	The investigation questions are defined. Analysts determine what is being investigated and what intelligence is needed.
Collection	Raw data is gathered from internal logs, threat feeds, OSINT, and commercial intelligence sources.
Processing	Raw data is sorted, organized, correlated, and converted into a usable format for analysis.
Analysis	Processed data is interpreted to answer the investigation questions and produce actionable intelligence.
Dissemination	Findings are shared with relevant stakeholders in the appropriate format for their role.
Feedback	Stakeholders provide feedback to refine future intelligence requirements. The cycle restarts.

Key CTI Frameworks and Standards

- **MITRE ATT&CK:** A knowledge base cataloguing adversary behavior by tactic and technique. Each technique has a unique ID (e.g., T1566 for Phishing). Used to label and communicate threat findings in a standard language recognized across the industry.
- **STIX (Structured Threat Information Expression):** A language that defines how threat intelligence objects (indicators, campaigns, TTPs, attack patterns) are represented and how they relate to each other.
- **TAXII (Trusted Automated eXchange of Indicator Information):** A protocol that defines how STIX-formatted threat intelligence is securely exchanged between organizations and automated systems in near real time.
- **Cyber Kill Chain:** A seven-phase adversary campaign model (Reconnaissance, Weaponization, Delivery, Exploitation, Installation, C2, Actions on Objectives). Used to identify which phase of an attack is observable in the current evidence.

DETECTION OPPORTUNITIES



- Based on the SIEM Alerts, the attacker (vipivillain@badbank.com) sent a phishing email to John Doe. The victim downloaded an executable file (flbpfu.exe), which modified registry files upon execution. The malware then established an outbound network connection to the attacker's external IP address (\$91.185.23.222\$). Following this activity, the attacker achieved privilege escalation, successfully logging into the network using the Administrator account.

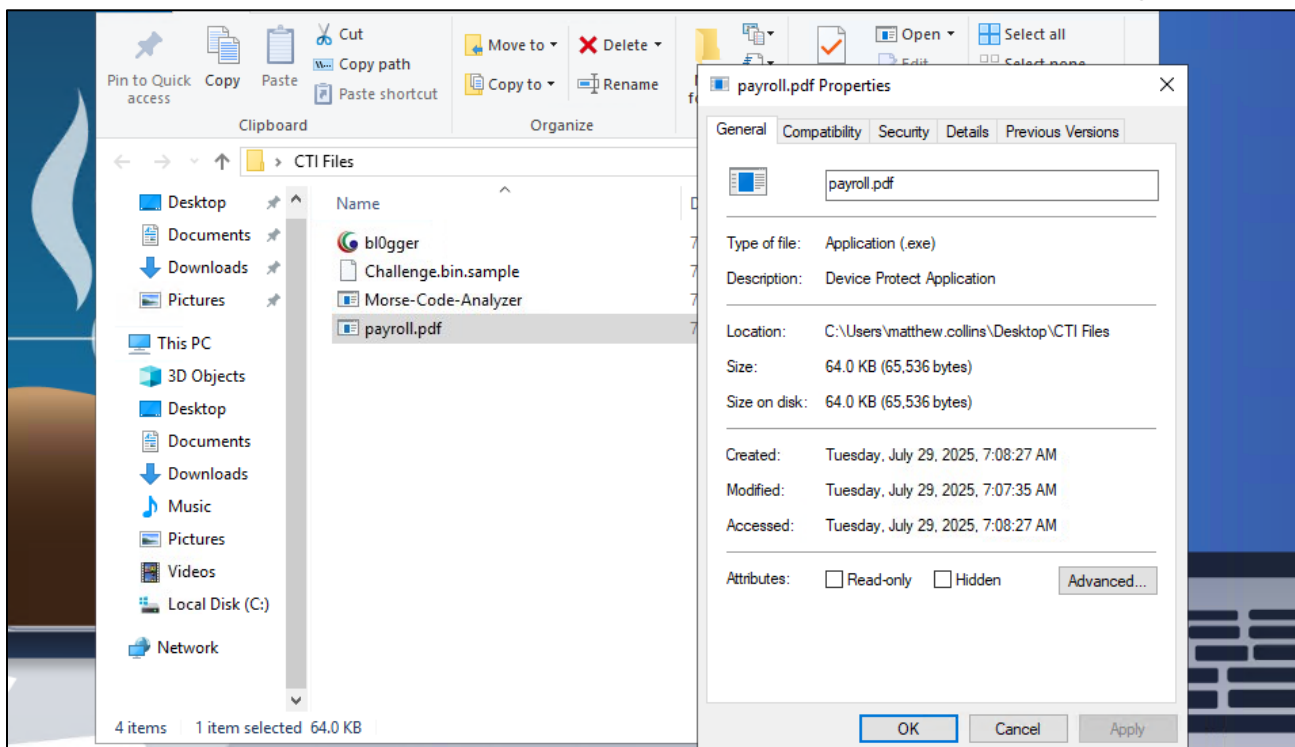
File and Hash Threat Intel

This room simulates a SOC scenario where multiple suspicious files are submitted for analysis. Attackers frequently disguise malicious files using misleading names, fake extensions, or by mimicking legitimate software. The investigation workflow covers detection of the disguise, hash extraction, and multi-platform threat intelligence enrichment.

File Masquerading and Double Extensions

Attackers commonly use two deception techniques to trick users and analysts. First, they name malicious executables after trusted programs (e.g., windows-update.exe, AdobeUpdate.exe) to pass casual inspection. Second, they use double extensions such as invoice.pdf.exe, where the operating system hides the true .exe extension by default, showing only invoice.pdf to the user.

Detection of double extensions requires either enabling file extension visibility in Windows Explorer or inspecting the file type directly using the file command or a tool like PESTudio. A file that appears to have a document extension but reports as a PE32 executable is a clear indicator of masquerading.



Hash Extraction

SHA256 hashes were extracted using PowerShell's Get-FileHash command and CMD. For the bl0gger.exe sample, the SHA256 hash was computed and submitted to Virus Total. For the Morse-Code-Analyzer.exe sample, the hash was formatted as sha256: hash and submitted to MalwareBazaar to compare vendor verdicts across platforms.

```
PS C:\Users\matthew.collins\Desktop\CTI Files> certutil -hashfile bl0gger.exe sha256
SHA256 hash of bl0gger.exe:
2672b6688d7b32a90f9153d2ff607d6801e6cbde61f509ed36d0450745998d58
CertUtil: -hashfile command completed successfully.
```

FILE HASH

File Overview

1f8806869616c18cbae9ffc581c0428915d32fb70119df16d08078d92d1a5e3

Malicious: 4

Suspicious: 1

Harmless: 0

Undetected: 59

Timeout: 6

Name: Morse-Code-Analyzer

Classification: trojan.dll hijack/generic

Size: 189,440 bytes

Type: Win32 EXE (.exe)

Submitted: 2025-04-28T09:14:33Z

Analysed: 2025-05-10T14:22:07Z

File Details

Basic Properties

Name: Morse-Code-Analyzer

SHA256: 1f8806869616c18cbae9ffc581c0428915d32fb70119df16d08078d92d1a5e3

Magic: PE32 executable (GUI) Intel 80386, for MS Windows

File Type: Win32 EXE

Size (Bytes): 189440

Submitted: 2025-04-28T09:14:33Z

Analysed: 2025-05-10T14:22:07Z

result Clean / Non-malicious

category benign

Kaspersky	HEUR:Trojan.Win32.Generic	trojan	Show more
ESET-NOD32	Win32/DLL.SideLoad	malware	Show more
Microsoft	Trojan:Win32/Loader	trojan	Show more
Sophos	Mal/DLLHijack-A	malware	Show more
TrendMicro	Undetected	undetected	Show more

Hybrid Analysis (bl0gger.exe)

bl0gger.exe was investigated on Hybrid Analysis to review dynamic behavior. The tags assigned to the sample on the platform were BlackMoon, Discovery, windows-server-utility. The process tree showed regsvr32 %WINDIR%\Media\ActiveX.ocx /s stealth command line were executed from the file and spawned a other process named werfault.exe.

Sandbox Analysis

Classification Tags (Hybrid Analysis)Process Tree

Registry ActivityTechnique Mapping

BlackMoon, Discovery, windows-server-utility

Malicious

85/100

Windows 10 64-bit

Analysis Duration: 120 seconds

Sandbox Analysis

Classification Tags (Hybrid Analysis)Process Tree

Registry ActivityTechnique Mapping

Parent Process: bl0gger.exe (PID 2148)

└─ regsvr32.exe (PID 3012) – ActiveX component registration

└─ werfault.exe (PID 3180) – Windows Error Reporting (child process)

Observed Command:

regsvr32 %WINDIR%\Media\ActiveX.ocx /s

Note:

The /s flag runs regsvr32 silently, suppressing dialog boxes. This is a Living-off-the-Land (LotL) technique using a trusted Windows binary.

Virus Total (Challenge.bin.sample)

An investigation into the dynamic behavior of Challenge.bin.sample via VirusTotal revealed that it is classified under the akira and filecryptor malware families. The sample was first seen in the wild on October 30, 2024, at 17:17:24 UTC.

Upon execution, the malicious file drops a ransom note titled akira_readme.txt. It then utilizes PowerShell to run Get-WmiObject Win32_Shadowcopy | Remove-WmiObject, a command designed to

delete volume shadow copies. According to the MITRE ATT&CK framework, this behavior maps directly to technique T1490: Inhibit System Recovery.

62
/ 71

Community Score

43b0ac119f957bb209d86ec206ea1ec3c51dd87beb7ba649c7ec73756e7
Challenge.bin.sample

Size
1.02 MB

Last Analysis Date
1 day ago

8
EXE

62/71 security vendors flagged this file as malicious

peexe 64bits detect-debug-environment checks-cpa-name calls-wmi long-sleeps

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 21

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat labelransomware.akira/filerecryptor

Threat categoriesransomware trojan worm

Family labelsakira filecryptor misc

Security vendors' analysis

Do you want to automate checks?

AhnLab-V3	Trojan.Win.Generic.R651.893	Alibaba	Ransom.Win64/Akira_d44ffb24
Alibaba	Ransomware.Win/Akira.A	ALYac	Trojan.Ransom.Filerecoder
Antiy-AVL	Trojan(Ransom)/Win64.Akira	Arcabit	Trojan.Ransom.Akira.26
Arctic Wolf	Unsafe	Avast	Win64:MalwareX-gen [Misc]

Shell Commands

"C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe" /svc

"C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s StorSvc

"C:\Windows\System32\svchost.exe -k NetworkService -p

"C:\Windows\System32\svchost.exe -k netsvc -p

"C:\Windows\System32\svchost.exe -k saprsv

"C:\Windows\system32\lsass.exe

"C:\Windows\system32\appsvcs.exe

"C:\Windows\system32\svchost.exe -k LocalService -s W32Time

"C:\Windows\system32\svchost.exe -k UninstallSvcGroup

"C:\Windows\system32\svsmbc.exe

"C:\Windows\system32\wbem\wmiiprsv.exe -secured -Embedding

powershell.exe -Command "Get-WmiObject Win32_Shadowcopy | Remove-WmiObject"

"%SAMPLEPATH%\43b0ac119f957bb209d86ec206ea1ec3c51dd87beb7ba649c7ec73756e7.exe"

"%SAMPLEPATH%\xy.exe"

"C:\Program Files\Google\1276_214657102\bin\updater.exe" --update --system --enable-logging --module="chrome/updater"~2 /sessionId [5BA74344-00EF-488E-9EA2-E8BASC18932C]

"C:\Program Files\Google\1516_1313231654\bin\updater.exe" --update --system --enable-logging --module="chrome/updater"~2 /sessionId [033ETC31-94DE-4CCT-A8B0-TEFA47A8F8]

"C:\Program Files\Google\1864_1600437160\bin\updater.exe" --update --system --enable-logging --module="chrome/updater"~2 /sessionId [C6B8BEAF-BD7F-4C2D-90FA-F161C0FF9C64]

"C:\Program Files\Google\2448_612056645\bin\updater.exe" --update --system --enable-logging --module="chrome/updater"~2 /sessionId [26239C8F-18EB-40C1-A07B-8FE39FA33E26]

"C:\Program Files\Google\2876_698020052\bin\updater.exe" --update --system --enable-logging --module="chrome/updater"~2 /sessionId [65350831-7400-4DDA-B3EC-34019354D850]

"C:\Program Files\Google\3094_205802623\bin\updater.exe" --update --system --enable-logging --module="chrome/updater"~2 /sessionId [B0727312-740F-690F-4E7B-CA7010183156]

DETECTION OPPORTUNITIES

- Alert on files with double extensions (e.g., pdf.exe) appearing in user download or temp directories.
- Automate hash enrichment on any new executable detected on an endpoint: submit to Virus Total and MalwareBazaar as the first triage step before analyst review.
- When Virus Total returns a MITRE Persistence technique in the Behavior tab, treat the file as high priority regardless of the overall detection count.

IP and Domain Threat Intel

In a SOC environment, alerts frequently contain nothing more than an IP address or a domain name. On their own these artefacts provide almost no context. Enrichment using OSINT platforms transforms them into a complete profile: who owns the infrastructure, what services it exposes, how long it has existed, and whether it has a history of malicious activity.

The scenario presented seven artefacts for triage: two suspicious domains from phishing emails and several IP addresses from outbound proxy logs. The objective was to enrich each one and recommend an action with a justification.

DNS and Domain Enrichment

The first flagged domain was purematrixa[.]com, an active command-and-control (C2) and payload delivery infrastructure used by threat actors to distribute malware. A DNS report was downloaded, and the A Record entry was examined. Multiple A records mapping to the same domain can indicate load balancing or CDN hosting. When the IP addresses in an A record belong to Cloudflare Infrastructure, the true origin server is obscured, which is a common tactic for both legitimate services and phishing campaigns.

DNS record types reviewed in this room: A Records map a domain to an IPv4 address. MX records identify the mail servers handling email for the domain. TXT records often contain SPF or DMARC policies for email authentication. CNAME records alias one domain name to another canonical domain.

Home > Whois Lookup > Purematrixa.com

Notice: Possible deprecation of Whois services after January 28, 2025. [More Info](#)

How does this work?

Whois Record for Purematrixa.com

Domain Profile

Registrar	Fewmoretape OU d/b/a Trustname.com IANA ID: 4318 URL: https://trustname.com/ Whois Server: https://whois.trustname.com/ abuse@trustname.com (p) +372.6884747
Registrar Status	ok
Dates	8 days old Created on 2026-05-31 Expires on 2027-05-31 Updated on 2026-05-31
Name Servers	DREW.NS.CLOUDFLARE.COM (has 44,973,303 domains) JOYCE.NS.CLOUDFLARE.COM (has 44,973,303 domains)
IP Address	104.21.91.69 - 1,156 other sites hosted on this server
IP Location	California - San Jose - Cloudflare Inc.
ASN	AS13335 CLOUDFLARENET - Cloudflare, Inc., US (registered Jul 14, 2010)
Domain Status	Registered And No Website
IP History	1 change on 1 unique IP addresses over 0 years
Hosting History	1 change on 2 unique name servers over 0 year

DomainTools Iris

The gold-standard internet intelligence platform

[Learn More](#)

[Preview the Full Domain Report](#)

Tools

Hosting History

Monitor Domain Properties

Reverse IP Address Lookup

Network Tools

Available TLDs

General TLDs

Country TLDs

The following domains are available through our preferred partners. Select domains below for more information. (3rd party site)

Taken domain.

Available domain.

Deleted previously owned domain.

IP Ownership and Registration

An infrastructure lookup was conducted on the malicious IP address 2[.158[.156[.150 using VirusTotal to extract ownership and registration data. The intelligence indicates that the IP resolves to the Netherlands and actively hosts a Remcos Command and Control (C2) server. Network attribution maps the IP to the 1337 Services GmbH Autonomous System (ASN). Furthermore, external routing verification via BGP.Tools confirmed this ASN is explicitly categorized under server hosting and Tor services.

15 / 91

Community Score

15/91 security vendors flagged this IP address as malicious

2.58.56.50 (2.58.56.0/24)

AS 220558 (1337 Services GmbH)

NL

Last Analysis Date: 21 minutes ago

[Reanalyze](#) [More](#)

DETECTION

DETAILS

RELATIONS

COMMUNITY 5

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Basic Properties

Network	2.58.56.0/24
Autonomous System Number	220558
Autonomous System Label	1337 Services GmbH
Regional Internet Registry	RIPE NCC
Country	NL
Continent	EU

Last HTTPS Certificate

JARM Fingerprint

71d404d020040b00043d43d020000d7c611cae37a98d775ecafba1b33ca523

1337 Services GmbH

AS Number **210558**
Website <https://rdp.sh>



Overview

Prefixes

Connectivity

Whois

IX

Registered on

28 Oct 2021 (4 years old)

Registered to

[de.1337service...](#) (ripe)

Network status

Active, Allocated under RIPE

Prefixes Originated

22 IPv4, 3 IPv6

Network type

Content

Upstreams

- [AS35133](#) - Eranium B.V.
- [AS19318](#) - InterServer, Inc
- [AS24875](#) - NovoServe B.V.
- [AS23470](#) - ReliableSite.Net LLC
- [AS201814](#) - MEVSPACE sp. z o.o.
- [AS3920](#) - PUSHPKT OU
- [AS399486](#) - Virtuo

Rankings

- #135 for [Estimated Eyeballs in Germany](#)
- #128 for [Unique Domains in Germany](#)

Tags:

Server Hosting

Tor services

Exposed Services (Censys)

An infrastructure query was conducted on IP address 64[.]89[.]160[.]44 using Censys, revealing a total of nine open ports. Remote Desktop Protocol (RDP) was identified as the primary exposed service on the host. Crucially, the scan detected an active AsyncRAT Command and Control (C2) server operating on one of the exposed ports, utilizing an SSL/TLS certificate with a validity period of 3,935 days.



Free Plan

Home

MY WORK

Collections

Search History

Tags

EXPLORE MODULES

Adversary Investigation

Critical Infrastructure

Search 64.89.160.44

64.89.160.44 • HOST

REMOTE_ACCESS

AS OF JUN 01, 2026 | 14:33 UTC

Summary

Reverse DNS

No Data

Forward DNS

No Data

OS

No Data

9 Total Services

135 / DCERPC

139 / NETBIOS

445 / SMB

1000 / UNKNOWN

3389 / RDP

7777 / UNKNOWN

27015 / VALVE

27152 / VALVE

27383 / VALVE

Home

MY WORK

Collections

Search History

Tags

EXPLORE MODULES

Adversary Investigation

Critical Infrastructure

Resources

Upgrade to Unlock More

UNKNOWN 1000 / TCP

LAST OBSERVED JUN 01, 2026 | 04:01 UTC

DetailsRaw DataJSON

TLS HANDSHAKE

Version SelectedUNKNOWN

CERTIFICATE

Fingerprintac1384eac8534926109bf9f500de32df344cb1e7ff37b8fb5fdfe946832cc050

Subject DNCN=AsyncRAT Server

Issuer DNCN=AsyncRAT Server

NamesAsyncRAT Server

Browser TrustX Untrusted

Ever Valid?No

Free Plan

Home

MY WORK

Collections

Search History

Tags

EXPLORE MODULES

Adversary Investigation

Critical Infrastructure

Resources

Upgrade to Unlock More

Search advanced protocols

More results per search

API & MCP access

DcRat

CERTIFICATE

Actions

Pivots

Summarize Certificate

never-trustedunexpireduntrusted

AS OF FEB 12, 2026 | 04:17 UTC

Summary

Subject DNCN=DcRat

Issuer DNCN=DcRat Server, OU=qwqdanchun, O=DcRat By qwqdanchun, L=SH, C=CN

Serial NumberHexDecimal0xc0336184ae5b90457a4cc122f95002cc3323d75b

Validity PeriodFEB 20, 2025NOV 30, 2035(3935 days, 0:00:00)

FingerprintSHA-256SHA-15aec4bc

Trust and Revocation

Browser	Status	Trusted Path	Revoked	Expired
Chrome	X Untrusted	No	No	No
Edge	X Untrusted	No	No	No
Firefox	X Untrusted	No	No	No
Safari	X Untrusted	No	No	No

An infrastructure investigation was conducted on the malicious domain raytracingengine.com utilizing Censys, nslookup, and whois queries. The domain resolves to IP address 35.188.105.97, which is hosted within the Google Cloud Platform (GCP) ecosystem. The malicious server is geographically located in the United States, running on a Canonical Linux operating system, with an initial creation date of February 21, 2026.

35.188.105.97

HOST

Actions

Summarize Host

REMOTE_ACCESS

AS OF JUN 01, 2026 15:51 UTC

Summary

Reverse DNSNo Data

Forward DNSprohomedevolvers.com, www.mycdn.moe, mycdn.moe, api.mycdn.moe

OSCanonical Linux

11 Total Services

22 / SSH

23 / UNKNOWN

25 / SMTP

80 / HTTP

81 / HTTP

82 / HTTP

443 / HTTP

2323 / UNKNOWN

8080 / HTTP

8081 / HTTP

8082 / HTTP

Network and Geolocation

WHOIS NetworkGoogle LLC
GOOGLE-CLOUD

WHOIS OrganizationGoogle LLC
GOOGL-2

Autonomous System35.188.96.0/20
GOOGLE-CLOUD-PLATFORM -
Google LLC (396982), US

Council Bluffs, United States (US)

who.is

WHOISRDAPDNSUptimeDomain EventsWebsite MonitoringLogin

RDAP Domain Lookup

Look up structured registration data for domains using the Registration Data Access Protocol

raytracingengine.com

RDAP Information

IP Address: 3.222.192.211

Whois

RDAP

DNS Records

Uptime

Diagnostics

History

Hide Contact Info

Refresh Data

The domain raytracingengine.com is registered. You can still try to buy it [here](#).

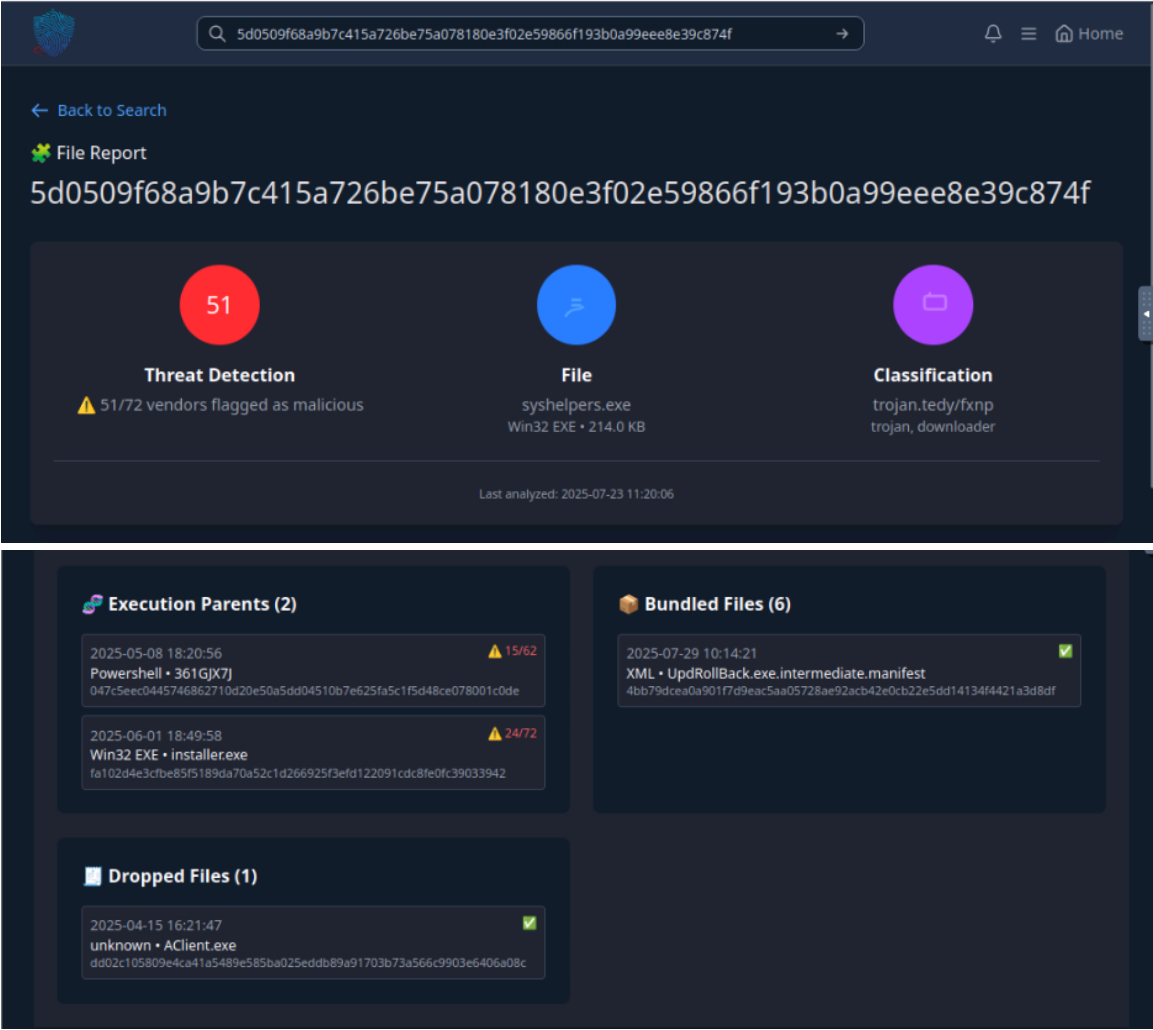
DETECTION OPPORTUNITIES

- Alert on outbound connections to domains that are newly registered (9 days old based on RDAP or WHOIS), as newly created domains are disproportionately associated with malicious campaigns.
- Alert on typo squatted domain lookups in DNS logs: compare resolved domains against a list of commonly impersonated software and brand names.
- When a Censys lookup shows a suspicious IP hosting 9 open ports with mixed service types, flag it for manual review as potential attacker infrastructure.

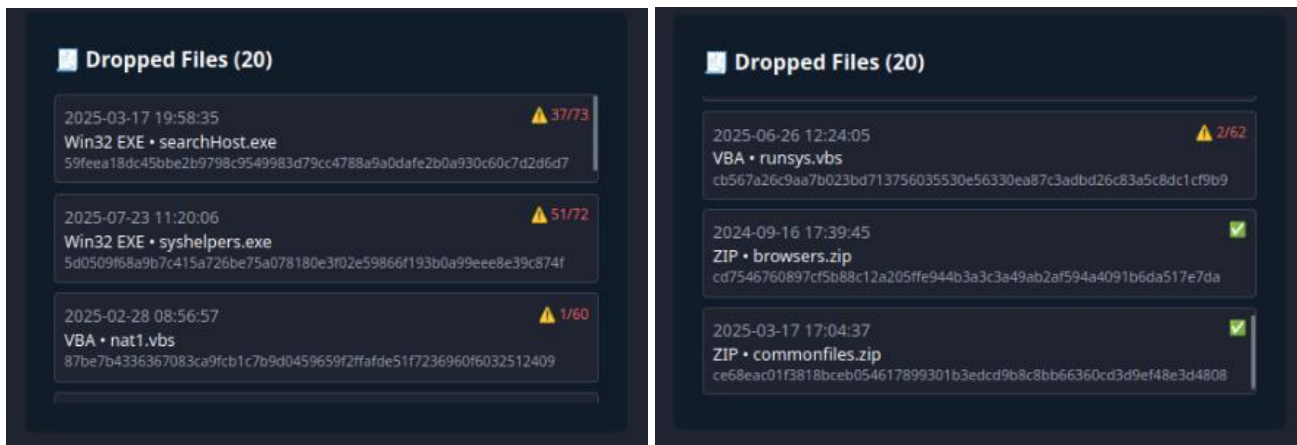
Artefact Investigation

During the initial triage, a Tier 1 analyst flagged two suspicious indicators for further investigation: IP address 101[.]99[.]76[.]120 and SHA256 hash 5d0509f68a9b7c415a726be75a078180e3f02e59866f193b0a99eee8e39c874f. To enrich these findings and distill them into actionable threat intelligence, an in-depth analysis was conducted using VirusTotal.

Analysis of the flagged SHA256 hash confirmed it is a Win32 executable associated with the execution parents 361GJX7J and installer.exe, which subsequently drops the file Aclient.exe.

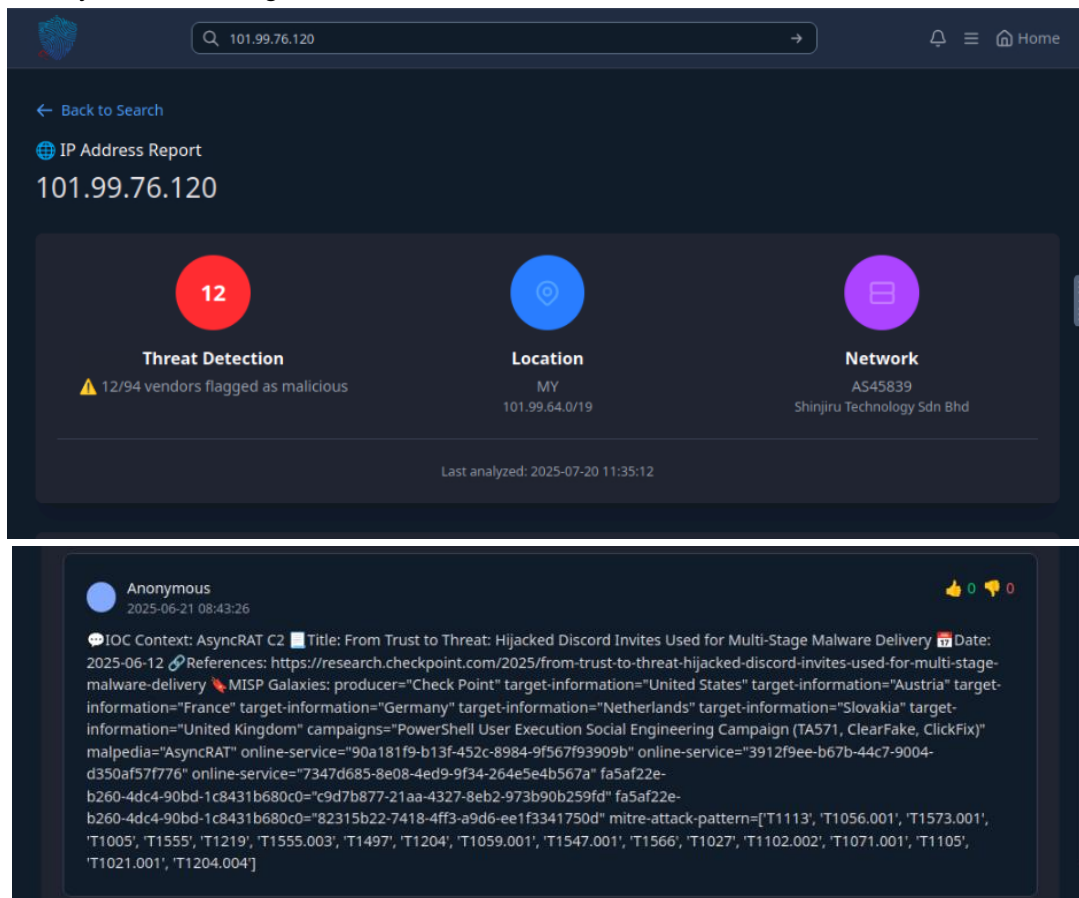


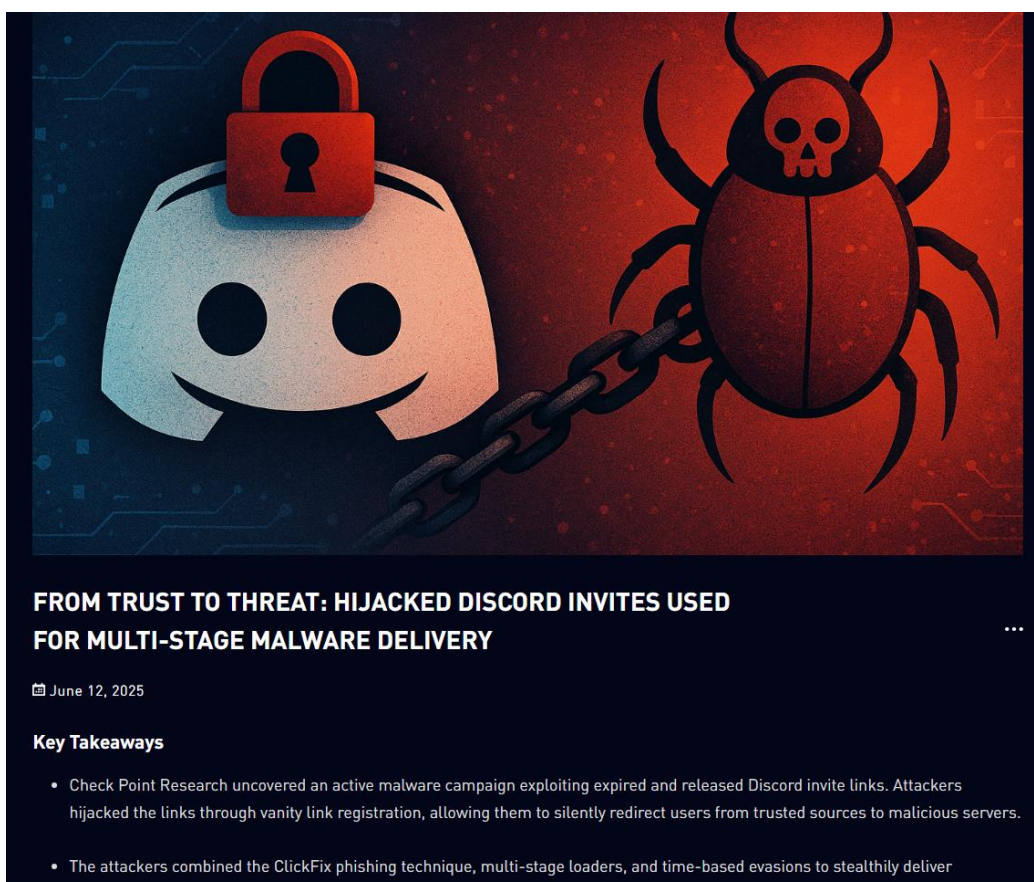
A deeper pivot into the telemetry of the installer.exe hash uncovered a secondary multi-stage infection chain responsible for dropping four additional malicious payloads: searchHost.exe, syshelpers.exe, nat.vbs, and runsys.vbs.



Pivoting to the network infrastructure investigation, analysis of the files communicating with the flagged IP address established a direct correlation to the AsyncRAT malware family, which is widely utilized for remote access and command-and-control (C2) operations.

Cross-referencing these network indicators led to a key threat intelligence report titled "From Trust to Threat: Hijacked Discord Invites Used for Multi-Stage Malware Delivery." According to the documentation, the campaign relied on compromised Discord invitations to redirect unsuspecting users to malicious delivery servers via the "ClickFix" phishing technique. Once the system was compromised, the threat actors deployed ChromeKats, a specialized credential-harvesting tool—to extract session cookies directly from the Google Chrome browser.





Investigation Methodology

The following workflow was applied across all enrichment investigations in this path.

Step	Action	Detail
1	Classify the Artefact	Identify whether the artefact is a file, hash, IP, or domain. This determines which enrichment tools to use first.
2	Extract or Confirm the Indicator	For files: compute SHA256 hash with Get-FileHash. For IPs and domains: defang the indicator (replace with [.]) before sharing.
3	Primary Enrichment	Submit to VirusTotal first. Check detection count, threat label, behavior tab, MITRE techniques, and community comments.
4	Secondary Enrichment	File or hash: cross-check on MalwareBazaar and submit to Hybrid Analysis. IP or domain: run RDAP, Shodan, Censys, crt.sh, and Cisco Talos.
5	Correlate Findings	Look for shared infrastructure: same TLS cert on multiple IPs, same ASN across multiple flagged IPs, same C2 domain in multiple sandbox reports.
6	Map and Document	Assign MITRE ATT&CK techniques to confirmed behaviors. Record all IOCs. Produce a verdict and recommended action for each artefact.

Key IOCs and Artefacts Investigated

IOC Type	Value	Context
Domain	purematrixa[.]com	An active command-and-control (C2) and payload delivery infrastructure used
IP Address	85[.]188[.]1[.]133	Flagged IP investigated via Shodan for exposed services and Censys for TLS certificate fingerprint.
IP Address	64[.]31[.]63[.]194	Investigated via RDAP. Registration date: 12/27/2010, 3:51:03 PM UTC. Roles: administrative and technical.
IP Address	166[.]11[.]160[.]118	Reputation investigated via Virus Total and Cisco Talos. Historical WHOIS attributed to Ace Data Centers, Inc.
TLS Fingerprint	5ea8e606...f056d12e	SHA256 TLS certificate fingerprint from Censys for IP 85[.]188[.]1[.]133 . Verified against crt.sh CT log entries.
Malware File	bl0gger.exe	Malicious file enriched on Virus Total and Hybrid Analysis. Tags and process tree extracted from sandbox report.
Malware File	Morse-Code-Analyzer.exe	Cross-referenced on Virus Total (MITRE persistence technique flagged) and MalwareBazaar (vendor verdicts compared).

Project Summary

This path built the complete threat intelligence enrichment skillset used in daily SOC operations. Each part added a new layer: Part 1 established the language and frameworks, Part 2 applied them to file and hash artefacts, and Part 3 applied them to network-based artefacts.

The most important lesson across all four parts is that no single platform provides the full picture. Virus Total may flag a file that MalwareBazaar classifies as clean. Shodan reveals open ports that a simple WHOIS lookup misses. crt.sh connects infrastructure that appears unrelated through a shared TLS certificate. Intelligence only becomes actionable when findings from multiple sources are correlated into a unified narrative.

Area Covered	Key Takeaway
CTI Foundations	Covered Data, Information, Intelligence distinction, four CTI classifications, six-phase lifecycle, and the MITRE ATT&CK, STIX, TAXII, and Cyber Kill Chain frameworks.
File and Hash	Detected file masquerading and double extensions. Used Get-FileHash, Virus Total, and Hybrid Analysis to enrich bl0gger.exe and Morse-Code-Analyzer.exe.
IP and Domain	Enriched 3 IPs and 1 typo squatted domain using RDAP, Shodan, Censys, and Virus Total. Correlated TLS fingerprint across platforms.
Invite Only	Capstone exercise applying the full enrichment workflow independently to a set of pre-flagged artefacts, producing a structured threat intelligence output.

Key Tools

Virus Total, Hybrid Analysis, RDAP, Shodan, Censys, Get-FileHash.

Core Skill

Multi-source enrichment and correlation: no single tool is sufficient. Intelligence is the result of synthesizing findings across all available platforms.
